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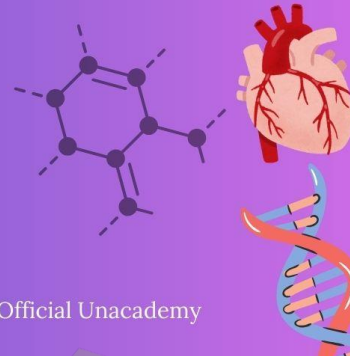
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LAST 10 YEARS

NEET PYQ



Chapter 11 : Photosynthesis in Higher Plants

1. Which of the following statements about RuBisCO is true? [NEET-2025]

- (1) It is active only in the dark
- (2) It has higher affinity for oxygen than carbon dioxide
- (3) It is an enzyme involved in the photolysis of water
- (4) It catalyzes the carboxylation of RuBP

2. Match List I with List II: [NEET-2025]

List-I	List-II
A. Chlorophyll a	I. Yellow-green
B. Chlorophyll b	II. Yellow
C. Xanthophylls	III. Blue-green
D. Carotenoids	IV. Yellow to Yellow-orange

Choose the option with all correct matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-I, B-II, C-IV, D-III
- (4) A-I, B-IV, C-III, D-II

3. Which of the following are required for the light reaction of Photosynthesis? [Re-NEET-2024]

- | | |
|---------------------|-------------------|
| A. CO ₂ | B. O ₂ |
| C. H ₂ O | D. Chlorophyll |
| E. Light | |

Choose the correct answer from the options given below:

- | | |
|------------------------|---------------------|
| (1) A, C, D and E only | (2) C, D and E only |
| (3) A and B only | (4) A, C and E only |

4. Pollen Which one of the following products diffuses out of the chloroplast during photosynthesis?

- (1) ADP (2) NADPH [Re-NEET-2024]
(3) O₂ (4) ATP

5. Which of the following are required for the dark reaction of photosynthesis? [NEET-2024]

- A. Light B. Chlorophyll
C. CO₂ D. ATP
E. NADPH

Choose the correct answer from the options given below:

- (1) A, B and C only (2) B, C and D only
(3) C, D and E only (4) D and E only

6. How many molecules of ATP and NADPH are required for every molecule of CO₂ fixed in the Calvin cycle? [NEET-2024]

- (1) 2 molecules of ATP and 3 molecules of NADPH
(2) 2 molecules of ATP and 2 molecules of NADPH
(3) 3 molecules of ATP and 3 molecules of NADPH
(4) 3 molecules of ATP and 2 molecules of NADPH

7. Given below are two statements: [NEET-2024]

Statement I: In C₃ plants, some O₂ binds to RuBisCO, hence CO₂ fixation is decreased.

Statement II: In C₄ plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

8. Match List-I with List-II [NEET-2023-MANIPUR]

List-I

List-II

- | | |
|-------------------|-----------------------------|
| (A) Chlorophyll a | (I) Yellow to yellow orange |
| (B) Chlorophyll b | (II) Yellow green |
| (C) Xanthophyll | (III) Blue green |
| (D) Carotenoid | (IV) Yellow |

Choose the correct answer from the options given below :

- (1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV) (2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(3) (A)-(III), (B)-(II), (C)-(IV), (D)-(I) (4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

9. Given below are two statements : [NEET-2023-MANIPUR]
- Statement I : RuBisCO is the most abundant enzyme in the world.
- Statement II : Photorespiration does not occur in C₄ plants.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- (1) Both Statement I and Statement II are correct
 - (2) Both Statement I and Statement II are incorrect
 - (3) Statement I is correct but Statement II is incorrect
 - (4) Statement I is incorrect but Statement II is correct
10. How many times decarboxylation occurs during each TCA cycle? [NEET-2023-MANIPUR]
- (1) Once
 - (2) Twice
 - (3) Thrice
 - (4) Many
11. Which out of the following statements is incorrect? [NEET-2023-MANIPUR]
- (1) Both ATP and NADPH + H⁺ are synthesised during non-cyclic photophosphorylation.
 - (2) Stroma lamellae lack PS II and NADP reductase
 - (3) Grana lamellae have both PS I and PS II
 - (4) Cyclic photophosphorylation involves both PS I and PS II
12. How many ATP and NADPH₂ are required for the synthesis of one molecule of Glucose during Calvin cycle? [NEET-2023]
- (1) 12 ATP and 12 NADPH₂
 - (2) 18 ATP and 12 NADPH₂
 - (3) 12 ATP and 16 NADPH₂
 - (4) 18 ATP and 16 NADPH₂
13. The reaction centre in PS II has an absorption maxima at [NEET-2023]
- (1) 680 nm
 - (2) 700 nm
 - (3) 660 nm
 - (4) 780 nm
14. Which of the following combinations is required for chemiosmosis? [NEET-2023]
- (1) Membrane, proton pump, proton gradient, ATP synthase
 - (2) Membrane, proton pump, proton gradient, NADP synthase
 - (3) Proton pump, electron gradient, ATP synthase
 - (4) Proton pump, electron gradient, NADP synthase

15. Given below are two statements :

[NEET-2022]

Statement I: The primary CO₂ acceptor in C₄ plants is phosphoenolpyruvate and is found in the mesophyll cells.

Statement II: Mesophyll cells of C₄ plants lack RuBisCo enzyme.

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct

16. Which one of the following is not true regarding the release of energy during ATP synthesis through chemiosmosis? It involves :

[NEET-2022]

- (1) Breakdown of electron gradient
- (2) Movement of protons across the membrane to the stroma
- (3) Reduction of NADP to NADPH₂ on the stroma side of the membrane
- (4) Breakdown of proton gradient

17. What is the role of large bundle sheath cells found around the vascular bundles in C₄ plants?

- (1) To increase the number of chloroplasts for the operation of Calvin cycle
- (2) To enable the plant to tolerate high temperature
- (3) To protect the vascular tissue from high light intensity
- (4) To provide the site for photorespiratory pathway

[NEET-2022]

18. When one CO₂ molecule is fixed as one molecule of triose phosphate, which of the following photochemically made, high energy chemical intermediates are used in the reduction phase ?

- (1) 1 ATP + 1 NADPH
- (2) 1 ATP + 2 NADPH
- (3) 2 ATP + 1 NADPH
- (4) 2 ATP + 2 NADPH

[Re-NEET-2022]

19. Match List - I with List - II :

[Re-NEET-2022]

List - I

List - II

- | | |
|---------------------------------|---|
| (a) Porins | (i) Pink coloured nodules |
| (b) leg haemoglobin | (ii) Lumen of thylakoid |
| (c) H ⁺ accumulation | (iii) Amphibolic pathway |
| (d) Respiration | (iv) Huge pores in outer membrane of mitochondria |

Choose the correct answer from the options given below :

- (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

20. Identify the correct statements regarding chemiosmotic hypothesis : [Re-NEET-2022]
- (a) Splitting of the water molecule takes place on the inner side of the membrane.
 - (b) Protons accumulate within the lumen of the thylakoids.
 - (c) Primary acceptor of electron transfers the electrons to an electron carrier.
 - (d) NADP reductase enzyme is located on the stroma side of the membrane.
 - (e) Protons increase in number in stroma. Choose the correct answer from the options given below:
- (1) (a), (b) and (e)
 - (2) (a), (b) and (d)
 - (3) (b), (c) and (d)
 - (4) (b), (c) and (e)
21. The first stable product of CO₂ fixation in sorghum is : [NEET-2021]
- (1) Pyruvic acid
 - (2) Oxaloacetic acid
 - (3) Succinic acid
 - (4) Phosphoglyceric acid
22. Which of the following statements is incorrect? [NEET-2021]
- (1) Both ATP and NADPH + H⁺ are synthesized during non-cyclic photophosphorylation.
 - (2) Stroma lamellae have PS I only and lack NADP reductase.
 - (3) Grana lamellae have both PS I and PS II.
 - (4) Cyclic photophosphorylation involves both PS I and PS II.
23. In the plastid, the site of dark and light reaction is [NEET-2021]
- (1) Grana and stroma respectively
 - (2) Grana and matrix respectively
 - (3) Matrix and grana respectively
 - (4) Stroma and grana respectively
24. The maximum absorption of light by chlorophyll occurs in which region of the absorption spectrum. [NEET-2021]
- (1) Red and green
 - (2) Yellow and green
 - (3) Blue and red
 - (4) Brown and red
25. Non-cyclic photophosphorylation results in the formation of [NEET-2021]
- (1) ATP and NADH + H⁺
 - (2) ATP only
 - (3) ATP and NADPH + H⁺
 - (4) ATP, ADP and NADH + H⁺
26. In light reaction, plastoquinone facilitates the transfer of electrons from : [NEET-2020]
- (1) PS-I to ATP synthase
 - (2) PS-II to Cytb₆-f complex
 - (3) Cytb₆-f complex to PS-I
 - (4) PS-I to NADP⁺

27. The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of
(1) 1 molecule of 4-C compound and 1 molecule of 2-C compound. [NEET-2020]
(2) 2 molecules of 3-C compound
(3) 1 molecule of 3-C compound
(4) 1 molecule of 6-C compound
28. During non-cyclic photophosphorylation, when electrons are lost from the reaction centre at PS II, what is the source which replaces these electrons? [NEET-2020 COVID-19]
(1) Oxygen (2) Water
(3) Carbon dioxide (4) Light
29. Which of the following statements is incorrect? [NEET-2020 COVID-19]
(1) RuBisCO is a bifunctional enzyme
(2) In C₄ plants, the site of RuBisCO activity is mesophyll cell
(3) The substrate molecule for RuBisCO activity is a 5-carbon compound
(4) RuBisCO action requires ATP and NADPH
30. In Hatch and Slack pathway, the primary CO₂ acceptor is: - [NEET-2019]
(1) Oxaloacetic acid (2) Phosphoglyceric acid
(3) Phosphoenol pyruvate (4) RuBisCO
31. One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the culture by splitting light through a prism. He observed that bacteria accumulated mainly in the region of:
(1) Violet and green light (2) Indigo and green light [NEET-2019]
(3) Orange and yellow light (4) Blue and red light
32. Oxygen is not produced during photosynthesis by: - [NEET-2018]
(1) Green Sulphur bacteria (2) *Nostoc*
(3) *Cycas* (4) *Chara*
33. Which of the following is not a product of light reaction of photosynthesis? [NEET-2018]
(1) ATP (2) NADH
(3) NADPH (4) Oxygen
34. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct? [NEET-2017]
(1) Increasing atmospheric CO₂ concentration up to 0.05% can enhance CO₂ fixation rate
(2) C₃ plants respond to higher temperatures with enhanced photosynthesis while C₄ plants have much lower temperature optimum
(3) Tomato is a greenhouse crop which can be grown in CO₂ -enriched atmosphere for higher yield
(4) Light saturation for CO₂ fixation occurs at 10% of full sunlight

35. Phosphonyl pyruvate (PEP) is the primary CO₂ acceptor in: [NEET-2017]
(1) C₄ plants (2) C₂ plants
(3) C₃ and C₄ plants (4) C₃ plants
36. Emerson's enhancement effect and red drop have been instrumental in the discovery of: - [NEET- I 2016]
(1) Photophosphorylation and non-cyclic electron transport
(2) Two photosystems operating simultaneously
(3) Photophosphorylation and cyclic electron transport
(4) Oxidative phosphorylation
37. In a chloroplast the highest number of protons are found in: - [NEET- I 2016]
(1) Stroma (2) Lumen of thylakoids
(3) Inter membrane space (4) Antennae complex
38. A plant in your garden avoids photorespiratory losses, has improved water use efficiency shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant? [NEET- I 2016]
(1) C₃ (2) C₄
(3) CAM (4) Nitrogen fixer
39. The process which makes major difference between C₃ and C₄ plants is: - [NEET- II 2016]
(1) Photorespiration (2) Respiration
(3) Glycolysis (4) Calvin cycle

ANSWER KEY

1. (4)
2. (2)
3. (2)
4. (3)
5. (3)
6. (4)
7. (3)
8. (3)
9. (1)
10. (2)
11. (4)
12. (2)
13. (1)
14. (1)
15. (4)
16. (1)
17. (1)
18. (1)
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27. (3)
28. (2)
29. (2)
30. (3)
31. (4)
32. (1)
33. (2)
34. (2)
35. (1)
36. (2)
37. (2)
38. (2)
39. (1)

SEEK PAKHUA